

Overlap Does Not Decide Independence

AP Statistics · Unit 2 · Topic 2.7 · B: 2026-10-29 · E: 2026-11-18 · TEACHER KEY

CED TETHER

- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- **EU VAR-4** — The likelihood of a random event can be quantified.
- **LO VAR-4.E** — Calculate probabilities for independent events and for the union of two events. [Skill 3.A]
- **EK VAR-4.E.1** — Events A and B are independent if, and only if, knowing whether event A has occurred (or will occur) does not change the probability that event B will occur.
- **EK VAR-4.E.2** — If, and only if, events A and B are independent, then $P(A | B) = P(A)$, $P(B | A) = P(B)$, and $P(A \cap B) = P(A) \cdot P(B)$.
- **EK VAR-4.E.3** — The probability that event A or event B (or both) will occur is the probability of the union of A and B, denoted $P(A \cup B)$.
- **EK VAR-4.E.4** — The addition rule states that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Essential question: Which probability comparison determines independence, and how is that different from checking overlap?

DOK-3 skill: 4.B · **FRQ pattern:** independent-vs-mutually-exclusive-decide-from-the-table

DOK rationale: Part (c) requires selecting and interpreting an independence check, separating independence from mutual exclusivity despite an overlapping sample, and tracing the multiplication-rule consequence of the decision.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Read both students' lines without endorsing the conclusion or the method.
- Begin the video follow-along at minute 5.
- Return to the table at minute 33. Circulate and ask students to name the two probabilities in every comparison.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Commit to an independence decision and cite one table feature.
- Complete the video follow-along about independence, intersections, and unions.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- What is the chance of instrument playing before you know pet ownership?
- What is that chance after you learn the student owns a pet?
- Could independent events occur together? What equation would they satisfy?

- For a complementary pair, what are the intersection and the product of the marginals?

Adult role

Solo teacher. Circulate 5 pairs. Require a probability equality or inequality for independence and a separate overlap statement for mutual exclusivity; do not accept the word related without both numbers.

[WARN] WATCH FOR

- Comparing $P(I | P)$ with $P(P)$ rather than $P(I)$.
- Using the product rule to manufacture an intersection before independence is established.
- Treating not mutually exclusive as a synonym for independent.

SCORING PART (c) — E / P / I

Expected elements

- **rejects-independence-with-probabilities** (required) — Rejects independence by comparing 0.70 conditional on pet ownership with the 0.50 marginal probability
- **separates-event-relationships** (required) — Explains that mutual exclusivity concerns overlap while independence concerns whether one probability changes given the other
- **gives-exclusive-dependent-pair** (required) — Uses owns a pet and does not own a pet, or an equivalent complementary pair, as mutually exclusive but not independent
- **states-product-consequence** (required) — Explains that independence would make the intersection equal 0.60 times 0.50, or 0.30, and contrasts it with 0.42

E	Rejects independence with the two required probabilities, distinguishes the two event relationships with a valid contextual pair, and correctly states the multiplication-rule consequence.
P	Makes the correct independence decision and probability comparison but incompletely explains mutual exclusivity, omits a valid pair, or does not connect independence to the product calculation.
I	Uses overlap alone to decide independence, says mutually exclusive means independent, or claims the actual intersection should equal 0.30 without treating that as a failed independence condition.

Common mistakes

- Comparing $P(I | P)$ with $P(P)$ instead of with $P(I)$
- Saying events must be independent whenever both can occur
- Calling owns a pet and plays an instrument mutually exclusive despite the count 84
- Using $P(P)P(I)$ for the actual intersection before independence has been established

[STAR] TODAY'S PROBLEM — annotated

Suppose a school surveys 200 students about whether they own at least one pet and whether they play a musical instrument. Let P be the event that a randomly selected surveyed student owns a pet and I the event that the student plays an instrument.

Aaliyah: *The events overlap because 84 students do both. Since they are not mutually exclusive, they must be independent.*

Marcus: *Overlap does not decide independence; the relevant probabilities must be compared.*

Pet ownership and instrument playing

Pet?	Plays	Does not	Total
Yes	84	36	120
No	16	64	80
Total	100	100	200

FIRST TAKE (5 min)

Do the events P and I seem independent? Cite one count or comparison from the table.

Key: Not graded. Aaliyah reaches the correct dependence decision from an invalid overlap rule; preserve that tension.

Rules callout “INDEPENDENCE, INTERSECTION, AND UNION (keep on the page)” is printed on the student sheet.

DOK 1 (a) State the counts for P , I , and $P \cap I$.

Key: There are 120 students in P , 100 in I , and 84 in $P \cap I$.

DOK 2 (b) Compute $P(I)$, $P(I | P)$, $P(P \cap I)$, and $P(P \cup I)$.

Key: $P(I) = 100/200 = 0.50$. $P(I | P) = 84/120 = 0.70$. $P(P \cap I) = 84/200 = 0.42$. By the addition rule, $P(P \cup I) = 0.60 + 0.50 - 0.42 = 0.68$ (equivalently, $136/200$).

DOK 3 (c) Decide whether pet ownership and instrument playing are independent for these students. Cite the two probabilities that settle it. Then evaluate Aaliyah’s reasoning: explain how mutually exclusive and independent events differ, give a pair of events from this context that is mutually exclusive but not independent, and state what independence would have allowed you to compute with the multiplication rule.

Key: The events are not independent because $P(I | P) = 0.70$ differs from $P(I) = 0.50$; knowing that a student owns a pet changes the modeled chance that the student plays an instrument. Aaliyah is right that P and I are not mutually exclusive, because 84 students are in both, but overlap neither proves nor disproves independence. Mutually exclusive events cannot occur together; independent events may overlap, but one event does not change the probability of the other. The events owns a pet and does not own a pet are mutually exclusive but not independent: their intersection has probability 0 while the product of their probabilities is $(0.60)(0.40) = 0.24$. If P and I were independent, the multiplication rule would give $P(P \cap I) = P(P)P(I) = (0.60)(0.50) = 0.30$; the actual value 0.42 provides another independence check.

EXIT

One thing the video changed about my first take: _____